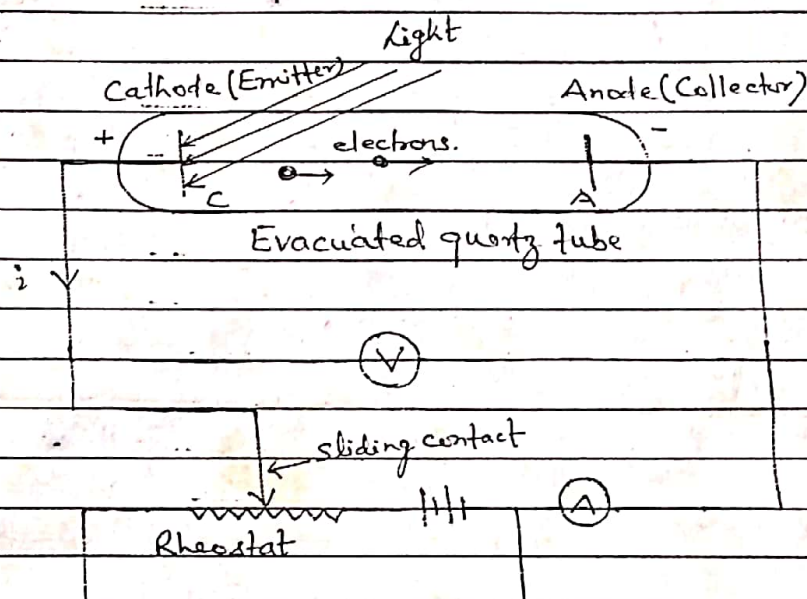


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✓ PHOTOELECTRIC EFFECT

The emission of electrons from a metal plate when illuminated by light or any other radiation of suitable wavelength (or frequency) is called photoelectric effect. The emitted electrons are called photoelectrons.

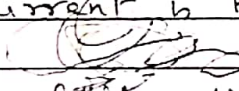


Fig(1)

The phenomenon was discovered by Heinrich Hertz in 1887 in the process of his research into electromagnetic waves. The phenomenon was experimentally verified by Hallwachs, Leonard, J. J. Thomson, R. A. Millikan & others. Afterwards it was discovered that alkali metals like lithium, sodium, potassium, rubidium and cesium eject electrons when visible light falls on them. Millikan investigated the effect over a wide range of light frequency and was given Nobel prize in 1923.

A simple experimental arrangement to study the photoelectric effect is shown above. Two metal plates C and A are sealed in an evacuated ^{quartz} tube. The tube is evacuated so that the electrons do not lose energy in collision with

molecules of the air. Light of reasonably high frequency passes through a transparent window in the wall of the tube and falls on the plate C, called the cathode or emitter. The electrons are emitted by C and collected by the plate A called the anode or the collector. The batteries and the rheostat (variable resistor) are used to produce and adjust the potential difference between C & A which is measured by the voltmeter V. The anode potential can be made positive or negative with respect to the cathode. The flow of electrons from emitter C to collector A is measured by the ammeter as a current i in the external circuit. The current is known as photocurrent.

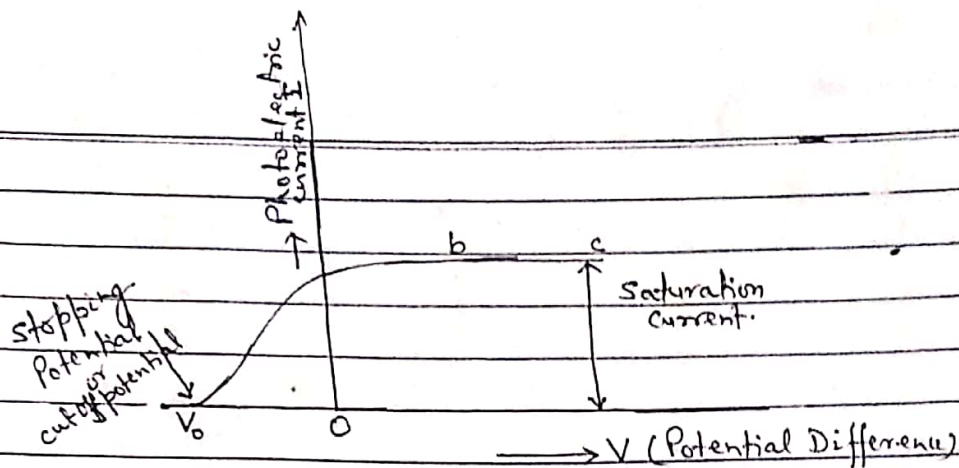
 It is observed that in the absence of light, there is no current. But when monochromatic light is allowed to fall on plate A, a current starts flowing in the circuit shown by ammeter. The number of electrons emitted and their kinetic energy depends upon the following factors: —

- (1) The potential difference between the two electrodes, i.e. between C & A.
- (2) The intensity of incident radiation.
- (3) The frequency of incident radiation.
- (4) The photo metal used.

1. The Effect Of Potential Difference : →

For a given photo metallic surface C, keeping the intensity & frequency of the incident radiation constant, let us consider the effect of potential difference between the two plates C & A.

As photoelectrons are emitted from C, they move towards the anode. At any time, the space between the cathode and the anode contains a number of electrons making up the space charge.



Fig(2)

This negative charge repels the fresh electrons coming from the cathode. However, some electrons are able to reach the anode & there is a photocurrent. (Here anode is at zero potential). When the anode is given a +ve potential with respect to cathode, the electrons are attracted towards the anode & therefore photocurrent increases. The current thus depends on the potential applied to the anode as shown by V-I graph.

If the potential is increased gradually, a situation arrives when the effect of space charge becomes negligible and therefore any electron that is emitted from cathode is collected by the collector A. The current then becomes constant. This constant current is called saturation current. Further increase in the potential hardly produces any appreciable increase in current. This is shown by the part bc in the V-I graph.

If the potential of the plate A is made -ve with respect to the cathode, the electrons are repelled by the anode. Only some of the photoelectrons having high kinetic energy reach the anode. They constitute the measured current. Thus the current decreases. When the retarding potential is further increased, there is further decrease in photocurrent and finally becomes zero at a particular potential value V_0 . It means at this potential the electrons having maximum kinetic energy are stopped from reaching to the collector. The values of -ve

* [The photo current does not immediately drop to zero but flows in the same direction as long as photoelectrons are emitted from the plate & ...]

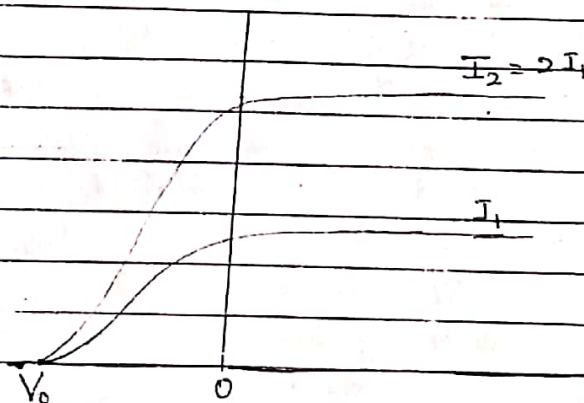
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potential of the plate A at which photocurrent just becomes zero is called the stopping potential. When $V = V_0$, the most energetic ejected electrons are turned back just before reaching the collector. As a photoelectron travels from the cathode to anode, the potential energy increases by eV_0 . This is equal to the decrease in the kinetic energy of the photoelectron. The kinetic energy of the fastest photoelectron as it reaches the anode is K_{max} , then

$$eV_0 = K_{max} \quad \text{--- (i)}$$

(2) Effect of intensity of incident radiation : $\rightarrow$

Now the intensity of incident radiation is increased from I to $2I$ and the above experiment is repeated. It is observed that the photoelectric current increases in the same ratio for all positive value of V i.e. The photoelectric current is also doubled. As V is made -ve, the photoelectric current decreases sharply and reaches zero at the same value of potential V_0 , the stopping potential.



When the intensity is doubled, the current is doubled, but the stopping potential V_0 remains same.

Fig(3) $\rightarrow$ Potential Difference

Hence it is concluded that - (i) the stopping potential is independent of the intensity of incident radiation. V_0 is related with K_{max} . In another words, the experiment shows that for light of a given frequency, K_{max} does

not depend on the intensity of the light source. Whether the source is dazzling bright or ~~feeble~~ dim, the maximum kinetic energy of the ejected electrons always has the same value.

(ii) The saturation current is proportional to the intensity of incident radiation. Higher is the intensity of incident radiation, higher is the saturation current.

(3) The effect of frequency of incident radiation: →

Now let us vary the frequency of the incident radiation while keeping the same emitting surface and same intensity of incident radiation. The stopping potential V_0 associated with each light of a given frequency. The graph between $\nu - V_0$ is plotted. ~~It is a straight line graph.~~ The graph shows that at ν_0 , the stopping potential is zero. The frequency ν_0 is known as threshold frequency and wavelength corresponding to ν_0 is called threshold wavelength.

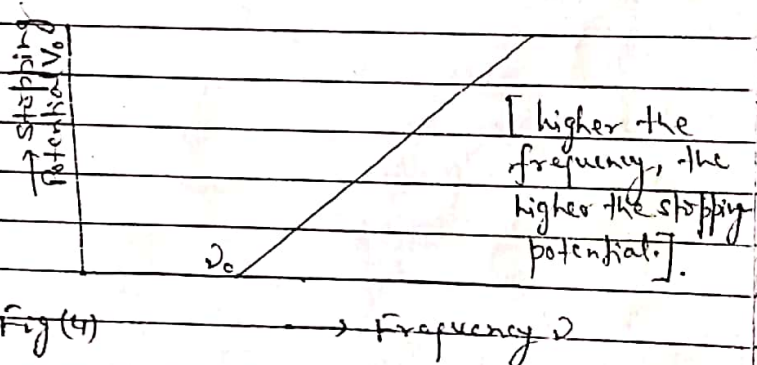
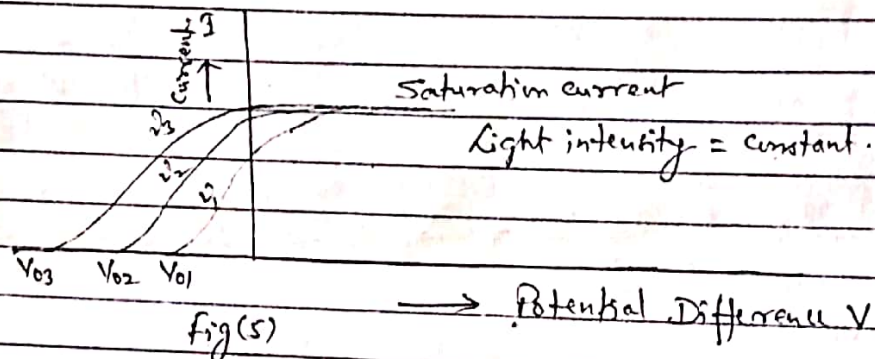


Fig (4)

The photoelectric effect occurs above this frequency while ceases below this frequency. The intensity of the incident light does not matter. For light below the threshold frequency, the photoelectric effect does not occur, no matter how bright the light source. Hence, the threshold frequency is defined as the minimum frequency (ν_0) of the incident radiation which can cause photoelectric emission. This frequency is just able to liberate electrons without giving them additional energy.

Threshold frequency can also be defined as the minimum value of frequency of incident light below which the photoelectric emission stops completely, no matter how ever the intensity of light may be.

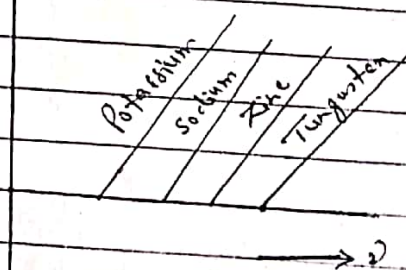
Several monochromatic radiations of frequency ν_1, ν_2, ν_3 etc are allowed to fall on the emitter. The intensity of all radiations having different frequencies is kept constant. In each case photoelectric current I for different values of ϕV is determined. A graph between $I-V$ for each frequency is obtained which shows that $V_{03} > V_{02} > V_{01}$ if $\nu_3 > \nu_2 > \nu_1$.



The observations indicate that retarding potential increased with increase in frequency. Since K.E of the photoelectrons $K_{max} = eV_0$, that in turn increases with increasing frequency. Thus, it is concluded that K.E is independent of the intensity of illumination but is directly proportional to the frequency of incident radiation.

(4) The effect of photo-metal: $\rightarrow$

If different photo-metals say potassium, sodium, zinc & tungsten are used and a curve between stopping potential and frequency is obtained plotted for each of them;



Fig(6)

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a family of curve is obtained. It is seen that all the curves have the same slope but different intersections on ν -axis which show that the value of threshold frequency is different for different metals. Thus observations indicate that the threshold frequency depends upon the nature of the photometal used in it. It is the characteristic of the metal.

LAWS OF PHOTO-ELECTRIC EMISSION: →

The experimental facts about photo-electric effect may be summarized as follows, which are known as the fundamental laws of photo-electric emission: -

1. The rate of photo-emission (the number of electrons emitted per second) i.e. the photoelectric current is directly proportional to the intensity of the incident radiations.
2. The velocity and hence the kinetic energy of emitted photoelectrons is independent of intensity.
3. The ^{max^m velocity or the} kinetic energy of photoelectron depends on the frequency of radiation & is directly proportional to the frequency of incident radiation.
4. The photo-emission takes place only above a certain frequency known as threshold frequency which is also characteristic of the kind of metal surface used in the experiment.
5. There is no time lag between the incidence of radiation and emission of photo-electron. The phenomenon is instantaneous.
6. For a given metal surface, the stopping potential V_0 is directly proportional to frequency but is independent of the intensity of incident light.